

Problem A. Multiplication Table

Time Limit 1000 ms

Mem Limit 524288 kB

Find the middle element when the numbers in an $n \times n$ multiplication table are sorted in increasing order. It is assumed that n is odd.

For example, the 3×3 multiplication table is as follows:

1	2	3
2	4	6
3	6	9

The numbers in increasing order are $[1, 2, 2, 3, 3, 4, 6, 6, 9]$, so the answer is 3.

Input

The only input line has an integer n .

Output

Print one integer: the answer to the task.

Constraints

- $1 \leq n < 10^6$

Example

Input	Output
3	3

Problem B. Reading Books

Time Limit 1000 ms

Mem Limit 524288 kB

There are n books, and Kotivalo and Justiina are going to read them all. For each book, you know the time it takes to read it.

They both read each book from beginning to end, and they cannot read a book at the same time. What is the minimum total time required?

Input

The first input line has an integer n : the number of books.

The second line has n integers $t_1, t_2, \dots, t_n$: the time required to read each book.

Output

Print one integer: the minimum total time.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq t_i \leq 10^9$

Example

Input	Output
3 2 8 3	16

Problem C. Powered Addition

Time Limit 1000 ms

Mem Limit 262144 kB

You have an array a of length n . For every positive integer x you are going to perform the following operation during the x -th second:

- Select some distinct indices $i_1, i_2, \dots, i_k$ which are between 1 and n inclusive, and add 2^{x-1} to each corresponding position of a . Formally, $a_{i_j} := a_{i_j} + 2^{x-1}$ for $j = 1, 2, \dots, k$. **Note that you are allowed to not select any indices at all.**

You have to make a nondecreasing as fast as possible. Find the smallest number T such that you can make the array nondecreasing after at most T seconds.

Array a is nondecreasing if and only if $a_1 \leq a_2 \leq \dots \leq a_n$.

You have to answer t independent test cases.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains single integer n ($1 \leq n \leq 10^5$) — the length of array a . It is guaranteed that the sum of values of n over all test cases in the input does not exceed 10^5 .

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$).

Output

For each test case, print the minimum number of seconds in which you can make a nondecreasing.

Examples

Input	Output
3 4 1 7 6 5 5 1 2 3 4 5 2 0 -4	2 0 3

Note

In the first test case, if you select indices 3, 4 at the 1-st second and 4 at the 2-nd second, then a will become $[1, 7, 7, 8]$. There are some other possible ways to make a nondecreasing in 2 seconds, but you can't do it faster.

In the second test case, a is already nondecreasing, so answer is 0.

In the third test case, if you do nothing at first 2 seconds and select index 2 at the 3-rd second, a will become $[0, 0]$.

Problem D. Chessboard and Queens

Time Limit 1000 ms
Mem Limit 524288 kB

Your task is to place eight queens on a chessboard so that no two queens are attacking each other. As an additional challenge, each square is either free or reserved, and you can only place queens on the free squares. However, the reserved squares do not prevent queens from attacking each other.

How many possible ways are there to place the queens?

Input

The input has eight lines, and each of them has eight characters. Each square is either free () or reserved ().

Output

Print one integer: the number of ways you can place the queens.

Example

Input	Output
<pre>.....*....**. ...*...</pre>	65